Dexterity: Listen to your *HEART*

ROUND 3: Dexterity National Finals

Situation

You are a Product Manager at XYZ Company. Your Chief Product Officer has approached you with a problem that was discussed in the chief executives' meeting. The company wants to enable pro-social behavior in the society via their application.

You are expected to design 3 features that enable this pro-social behavior. These features could be in the form of User Accessed features, or APIs that can be accessed by third party software that then drives pro-social behavior, or data that the vendor can analyze for insights that could identify pro-social behaviors.

What is pro-social behavior?

Prosocial behavior, or intent to benefit others, is a social behavior that "benefits[s] other people or society as a whole".

For Example: Swiggy rewarding customers with cashback points for returning the delivery bags.

The deliverables of the case include (but are not limited to):

- Problem Identification
- User and Market Research
- Prioritization of Features and proposing a brief road map
- Success metrics
- Mock-up and wireframes (not necessary but will be helpful)

Tasks to be done

Select the XYZ Company- It should be an existing and known product

Submit 5-8 slider presentation (excluding annexures, cover slide)

Bharatiya Vidya Bhavan's SPJIMR

OJAS XV
Graffiti of Glory

Dexterity: Listen to your HEART

Round 3: National Finals

INDIAN INSTITUTE OF MANAGEMENT KOZHIKODE
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Team : SOCRATES
College : IIM Kozhikode

Om Shubham

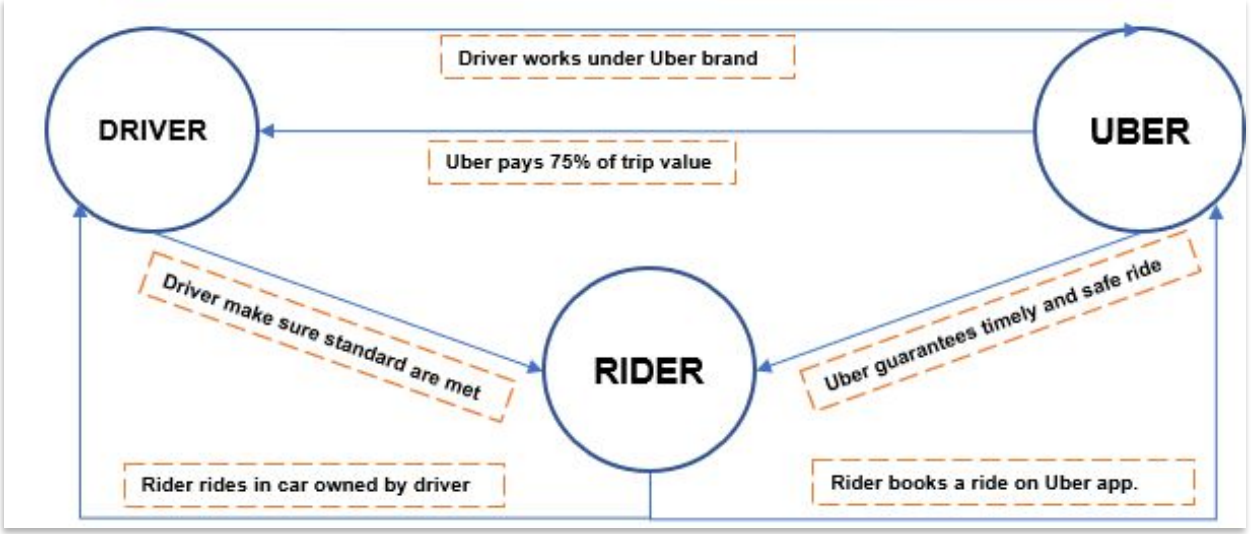
Problem Identification

Assumption: Uber platform for rides is considered for the case analysis and feature recommendation.

Uber is an on-demand cab aggregator that operates on a smartphone application and lets you book a cab to get from point A to point B, pre calculating the fare, estimating the time of arrival, and offering an option to split the cost with co-riders, all with a few taps on the app. The company disrupted the entire cab industry in 2008 when the founders were not able to find a cab on a cold winter evening. It all started with a simple question – "What if you could request a ride from your phone?" Today, the company is estimated to have over 118 million monthly active users worldwide, is a unicorn valued at ~$53 billion, and provides ride options from affordable bikes and scooters to Uber Air.

Services provided by Uber rides and its business model:

- ✔ UberX: The basic Uber ride.
- ✔ Bikes/Scooters: On-demand electric bikes/scooters.
- ✔ Uber Black: Luxury cars.
- ✔ UberXL: Extended UberX
- ✔ Uber Pool: Shared ride.
- ✔ Uber Comfort: New cars with extra legroom.
- ✔ Uber Green: Electric vehicles.



What is a Pro-Social Behaviour?

A social action that benefits or intents to benefit other individuals or society as a whole, such as assisting, sharing, giving, cooperating, and volunteering, is known as prosocial conduct. Prosocial activities also include following the law and acting in accordance with social norms.

Different types of Pro-social Behaviour

Helping
- Helping behaviours are a type of prosocial behaviour that benefits both individuals and society as a whole

Donating
- Charity is a wonderful form of prosocial behaviour. You can donate a variety of things to people who are less privileged

Volunteering
- Volunteering requires contribution of time, effort and talent to fulfill a need or benefit others without being forced to do it

Cooperating
- Cooperating is simply working together with one or more people to accomplish a common goal

Eco Friendly Behaviour
- Being eco friendly means living in a way that is not harmful to the environment

Complying With Social Norms
- Tipping a server at a restaurant
- Using good etiquette (table manners, saying please and thanks, etc.)

Market/User Research

Types of User Personas for Uber rides platform

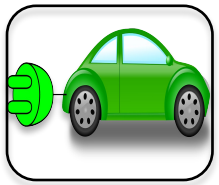
Use Cases/ Pain Points

Bio: John is a comfort user and books cars even for short trips where two wheeler vehicles can be used. This behaviour contributes in fuel wastage and traffic congestions

Pain Point: John has money to afford car travels even for short trips. He doesn't have any motivation to book scooters and bikes to compromise his comfort.

Bio: Alex is a comfort users who can afford booking cabs without any need to use car pooling services

Pain Point: Alex doesn't have any motivation to car pool for its short as well as long trips.



Bio: Bob is very sensitive about saving environment and take great care for the sake of it.

Pain Point: Bob doesn't know to which ride options on the platform are more eco-friendly than others.

Bio: Both Chetan- a comfort user and Shyam- a cost optimizer don't care to donate.

Pain Point: Chetan and Shyam don't have any incentive or motivation to donate for a good cause.

Comfort/Premium Users

- These users prefer comfort over money and can be divided based on these ride types:
- **Short Trip/Intracity Travels:** Users travelling within the city
- **Long Trip/Intercity:** Comfort users travelling across cities
- **Rentals:** Comfort users renting vehicles for a time period.

Bio: Ramesh is a driver on Uber who regularly uses the application to find rides and make money. He is provided with traffic and congestion information but doesn't know how to use it to make more eco friendly decisions.

Pain Point: Current Uber ride platform doesn't provide insights to help making eco friendly decisions.

The Cost Optimizers

- These users want to save money on each trip and can be divided as follows :
- **Short Trip/Intracity Travels:** Users optimizing travels within the city.
- **Long Trip/Intercity:** Users optimizing travels across cities
- **Rentals:** Users optimizing to rent vehicles for a time period.

Bio: Karan is a driver on uber platforms who ended up on a non operational road which was being shown as operational on the app. He wants to help other drivers.

Pain Point: Karan doesn't know how to inform everyone. He can just inform his friends.

Bio: Jack remains unsure of his travel plans most of the time, which has led him to cancel such plans multiple times.

Pain Point: Cancelling rides contributes to fuel wastage. Jack doesn't know about these repercussions, but being a comfort user he doesn't care about them either

Earners/Drivers

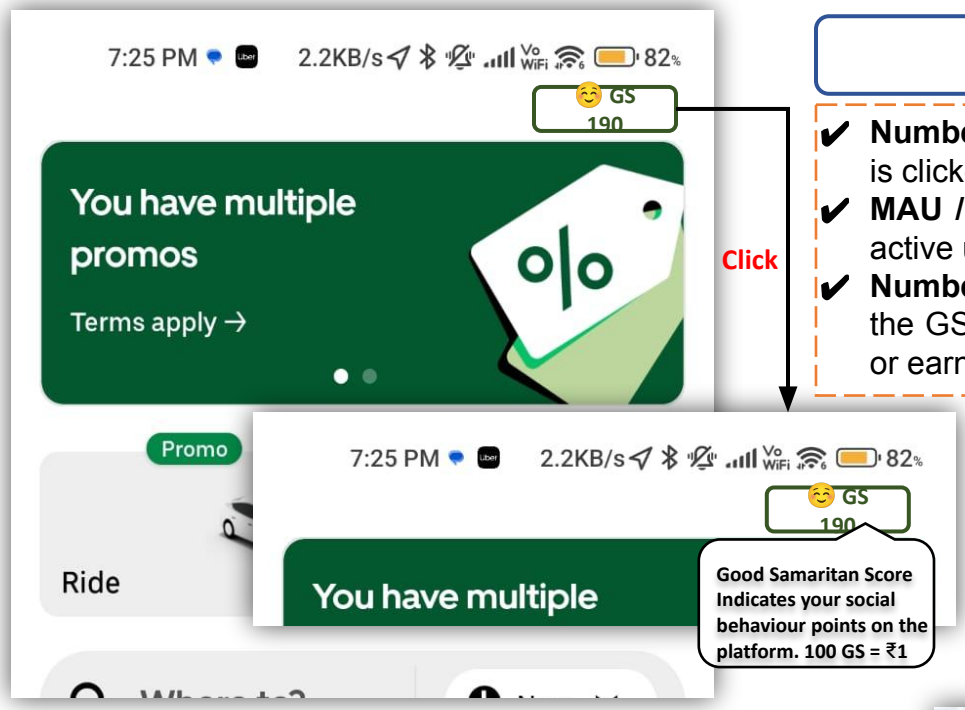
- These users earn money by using their own car and providing service to end customers through these types of trips:
- **Short Trip/Intracity Travels:** Drivers providing service within the city.
- **Long Trip/Intercity:** Drivers providing service across cities

Listing Solutions/Features

1. Good Samaritan Score for social features

Good Samaritan Score: Points accumulated for exhibiting pro-social activity on the platform- These points when accumulated up to a certain amount can be converted to cash and credited to 'Uber Cash' wallet. Example 100 GS = ₹ 1

- This score can be displayed for both riders and drivers using the platform.
- More points can be allocated according to the level of activity, points can also be reduced for showing anti social behaviour.

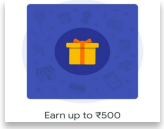


Success Metrics for the feature

- ✔ **Number of taps on score:** Number of times score bar is clicked.
- ✔ **MAU / DAU:** Measuring change in monthly and daily active users
- ✔ **Number of times points encashed:** Measures times the GS points used to gain cashback on rides booked or earn income (for drivers).
- ✔ **Retention rate:** Change in retention rate of the home page where GS score board is present.
- ✔ **Bounce rate:** Change in bounce rate of the home page where GS score board is present.

2. Perks for using electric vehicles

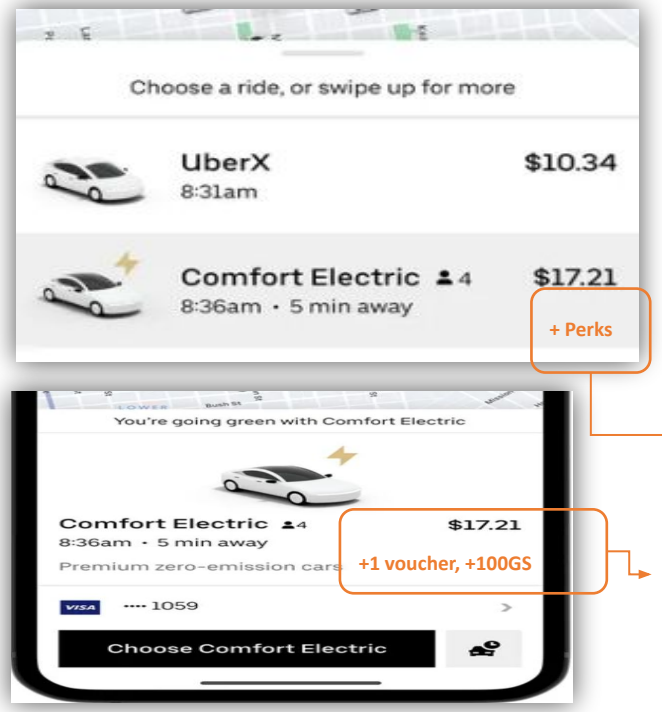
E-vehicles are more environment friendly than fuel driven vehicles. Extra perks could be given to customers in terms of cashback (that can be availed in next ride),if they choose to ride in E-vehicles.



- Perks in terms of vouchers can be given.
- Vouchers can be given in the form of scratch cards.
- GS score if incorporated can also be disbursed for choosing electric vehicles over other vehicles.

Success Metrics for the feature

- ✔ **Change in booking of electric vehicles:** Measure of change in number of booking of electric vehicles after feature is launched.
- ✔ **Number of scratch cards/vouchers availed:** Measure of number of scratch cards and vouchers availed once collected.
- ✔ **Number of new Electric car owners** : Measure of electric vehicle car owners who joined the platform to provide cab services.
- ✔ **Change in metrics of GS score**: Measure of change in metrics of GS score, if already incorporated.
- ✔ **Amount of Cashbacks availed**: Measure of amount of cashbacks availed after introducing this feature.

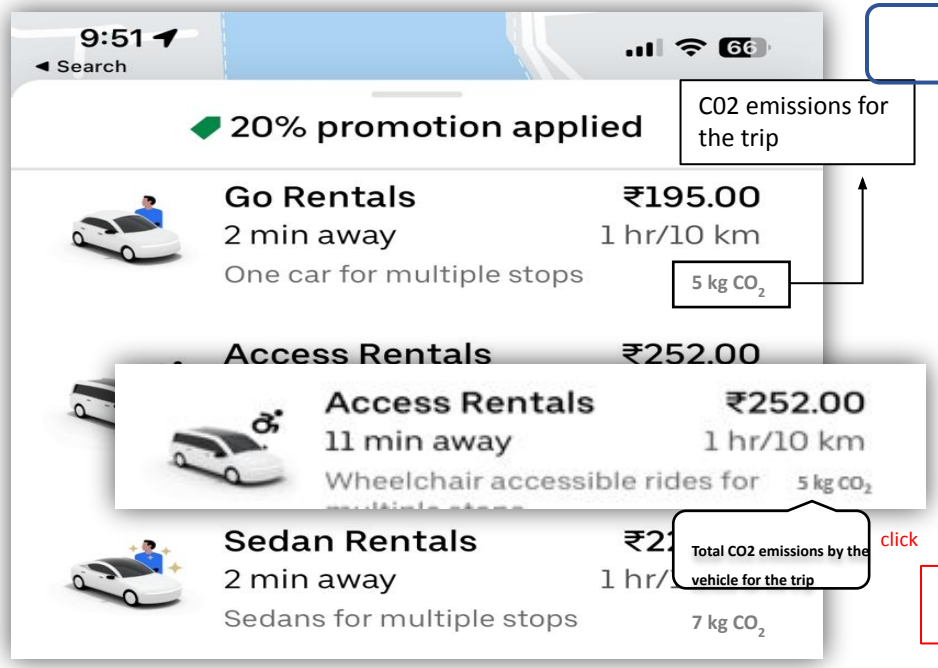


Listing Solutions/Features

3. Showing CO2 emissions by each vehicle

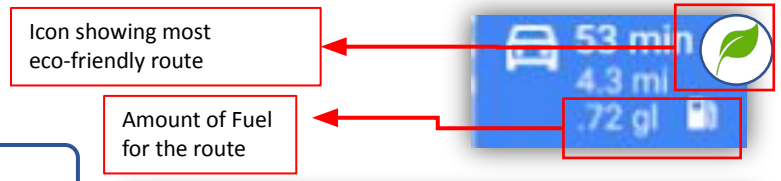
Google flight search services started showing approximate CO2 emissions by each flight. Same feature can be incorporated in Uber platform showing total CO2 emissions

- This is to encourage pro social behaviour and help individuals to take responsible decisions.
- GS score if incorporated can be used as a perk for this feature. More GS score should be offered for trips with less emission for the trip.



Success Metrics for the feature

- ✔ **Number of taps on emission information:** number of times CO2 bar information is clicked.
- ✔ **Retention rate for Cab information dashboard:** Total change in retention rate of the cab details dashboard after feature is incorporated.
- ✔ **Change in decision for better emission level:** Times user took eco friendly decision instead of cost optimization.



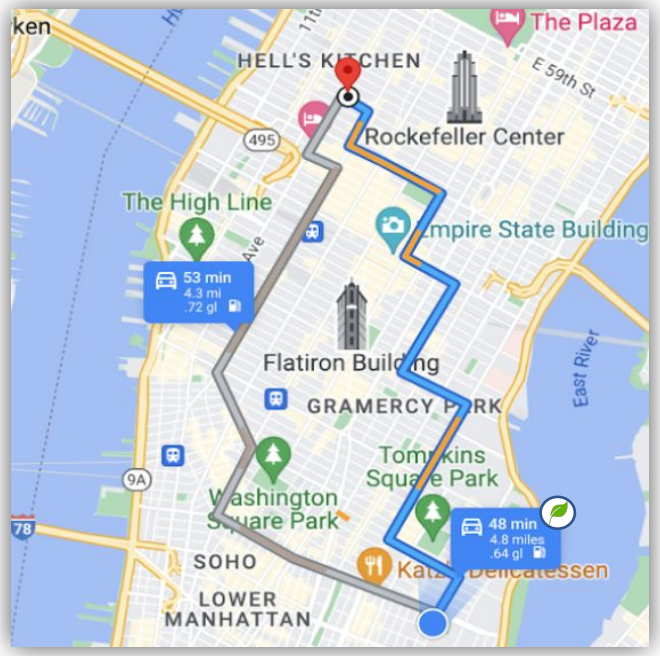
4. Configure eco-friendly routes for Drivers using Google API

- **Google Routes API:** By default, the Routes API returns the default route, meaning a route without factoring in fuel or energy-efficiency. When you enable eco-friendly routing, you still get the default route in response. In addition, you also get back an eco-friendly route showing the most fuel or energy efficient route based on your vehicle's engine type.

- By integrating this API in uber platform, fuel efficient route can be suggested to drivers.
- Drivers need to enter the vehicle engine which will be used as input to the routes API.
- Interface can consist of net fuel usage or power in case of e-vehicle to indicate how much fuel is saved by using alternate routes. Use of eco friendly marks on the fuel efficient path other than the default path.

Success Metrics for the feature

- ✔ **Total times API accessed:** Measure the total count the API is accessed to show the fuel efficient routes.
- ✔ **Total times fuel efficient route used**: Measure the number of times eco friendly route used and not the default routes.
- ✔ **Estimated Fuel saved:** Amount of fuel saved calculated from the suggestions showed through the API.
- ✔ **Estimated Cost of fuel saved:** Total cost saved by the driver according to the vehicle engine type. It is equal to estimated cost of fuel times cost per litre/gallon of that fuel.

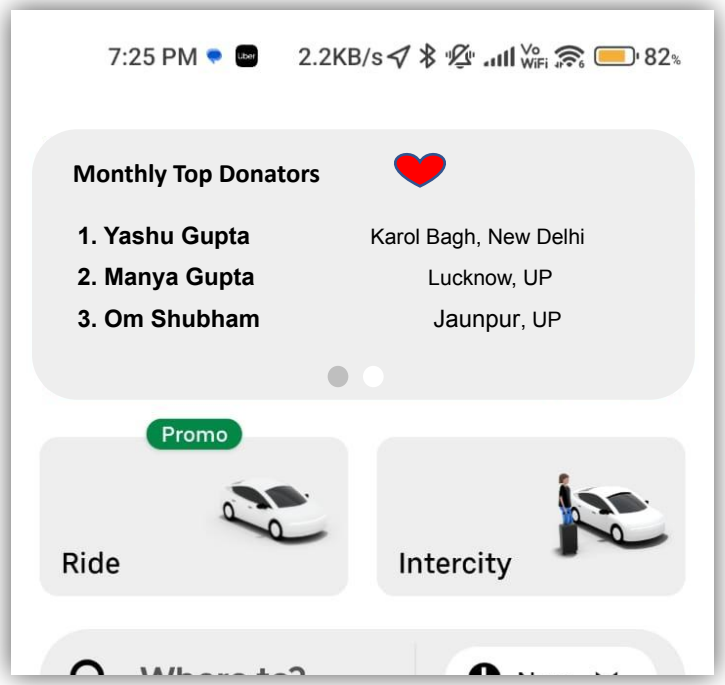


Listing Solutions/Features

5. Donate button/ Leader board

- **Donate Button:** Feature to donate money to non-profit organizations working across country for providing medical ambulance services to public.

- **Donation Leader board:** Feature to show monthly top individuals who donated highest total for the month.
- Extra perks can be given for the top donator in terms of one free premium ride or cashbacks. High GS score point can be also be credited if that feature is incorporated.



Success Metrics for the feature

- ✔ **Donation Amount:** Measure of total daily/monthly amount collected as donation fund for helping non profit organizations
- ✔ **Retention rate:** Change in retention rate of the home page where Donation Leader board is present.
- ✔ **Conversion rate:** This is the percentage of users who actually make a donation after clicking on the donate button
- ✔ **Perks disbursed for top donators:** Measure of total value of perks donated to top donators if the feature is incorporated.
- ✔ **Repeat donation rate:** This is the percentage of users who make more than one donation.

6. Non operational Route alert by drivers

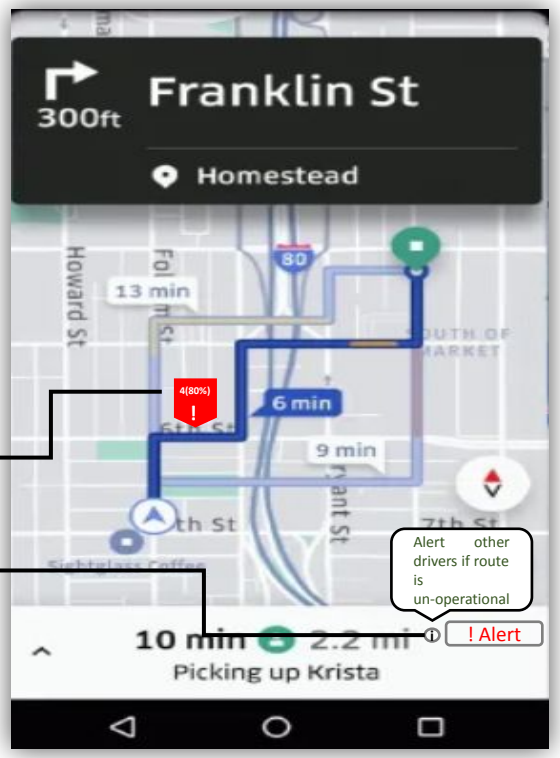
- Many of the times map shows or leads drivers to non operational routes. Drivers don't have any feature to alert other driver who are nearby.
- The interface will give drivers the option to click an alert button if they face a non operational route showing active on the map.
- Other drivers will be able to see a mark on the route and the number of drivers reported the route. Driver will decide whether they want to proceed with the current route or deviate to the other route.
- If the driver alerting for non operational route passes through that route then the alert by the driver will be recalled assuming it was a false alert.
- False alert also can be notified if driver reaches the alert mark location. Confidence rate will be calculated based on alert and false alerts.

Success Metrics for the feature

- ✔ **Number of times information button clicked:** Measure of number of times information button is clicked, to know more about the alert features or about the alert marker by the drivers.
- ✔ **Number of times Alert button used:** Measure total number of times alert button clicked to report no operational routes.
- ✔ **Number of times drivers deviated:** Measure of total number of times a driver deviated from current route if there is a alert mark shown in current path.
- ✔ **Number of false alert reported:** Measure of number of times false alerts notified.

Alert marker showing number of drivers alerted and confidence rate.

4(80%) !



Alert and Information buttons

Features Prioritization

Using RICE framework for prioritization

Feature	Reach	Impact	Confidence	Effort	Rice Score
1. Good Samaritan Score for social features	9	4	80	3	9.6
2. Perks for using electric vehicles	6	3.5	60	3	4.2
3. Showing CO2 emissions by each vehicle	5	2	70	1	7.0
4. Configure eco-friendly routes for Drivers using Google API	5	4	80	2	8.0
5. Donate button/ Leader board	6	3	75	3	4.5
6. Non operational Route alert by drivers	4	2	65	2	2.6

We will calculate the RICE score as follows:

$$\frac{\text{Reach} \times \text{Impact} \times \text{Confidence}}{\text{Effort}} = \text{RICE SCORE}$$

Assumptions – Reach max score is 10 points, Impact and effort max score is 5 points and Confidence is in percentage terms (out of 100).

Recommendations/Summary

RICE score of the features in descending order comes out to be:

1. Good Samaritan Score for social feature (Score: 9.6)
2. Configure eco-friendly routes for Drivers using Google API (Score: 8)
3. Showing CO2 emissions by each vehicle (Score:7)
4. Donate button/ Leader board (Score: 4.5)
5. Perks for using electric vehicles (Score: 4.2)
6. Non operational Route alert by drivers (Score: 2.6)

#1. Good Samaritan Score for social features

#2 Configure eco-friendly routes for Drivers using Google API

#3. Showing CO2 emissions by each vehicle

- The features if implemented should be rolled and tested out as in above sequential order. Further other features can also be integrated with GS score by allocating GS points in order of impact of the pro-social behaviour.
- Eco-friendly routes feature even though has low reach because the feature is specifically for drivers, it will have high impact in terms of low fuel wastage and low power consumption in electric vehicle case.
- Showing CO2 emissions feature doesn't have much impact as most optimizer user will give preference to money saved rather than emission level of the trip whereas comfort users will mostly go for saving time with having greater comfort. Effort required to build this feature is low due to which it can be prioritized to implement earlier.

Thank You !